

L3 LECTURE NOTES

WEEK 2

INFINITE SEQUENCES AND SERIES

SEQUENCES

A sequence is a list of numbers $a_1, a_2, a_3, \dots, a_n, \dots$ in a given order.

Each of a_1, a_2, a_3 and so on represents a number. These are the terms of the sequence.

$2, 4, 6, 8, \dots, 2n, \dots$

$$\begin{aligned} f: \mathbb{N} &\rightarrow \mathbb{R} \\ n &\mapsto a_n = f(n) \end{aligned}$$

has first term $a_1 = 2$, second term $a_2 = 4$ and n th term $a_n = 2n$. The integer n is called the index of a_n , and indicates where a_n occurs in the list. Order is important. The sequence $2, 4, 6, 8, \dots$ is not same as the sequence $4, 2, 6, 8, \dots$

Representing Sequences

→ we can think of the sequence

$$a_1, a_2, a_3, \dots, a_n, \dots$$

as a function sends 1 to a_1 , 2 to a_2 , 3 to a_3 , ...

and in general sends the positive integer n to the n th term a_n . More precisely, an infinite sequence of numbers is a function whose domain is the set of positive integers.

→ The function associated with the sequence $2, 4, 6, \dots, 10, 12, \dots, 2n$, sends 1 to $a_1 = 2$, 2 to $a_2 = 4$ and so on $a_n = 2n$.

→ we can equally well make the domain the integers larger than a given number n_0 , and we allow sequences of this type also. For example, the sequence

$$12, 14, 16, 18, \dots$$

$$a_n = 10 + 2n \quad n = 1, 2, \dots$$

also we can represent it in the form $b_n = 2n \quad n = 6, 7, \dots$

→ Sequences can be described by writing rules that specify their terms, such as

$$a_n = \sqrt{n} \quad b_n = (-1)^{n+1} \frac{1}{n} \quad c_n = \frac{n-1}{n} \quad d_n = (-1)^{n+1}$$

$$\{a_n\} = \{\sqrt{1}, \sqrt{2}, \sqrt{3}, \dots, \sqrt{n}, \dots\}$$

another notation $\{a_n\} = \{\sqrt{n}\}_{n=1}^{\infty}$

$$\{b_n\} = \left\{ 1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots, (-1)^{n+1} \frac{1}{n}, \dots \right\}$$

$$\{c_n\} = \left\{ 0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots, \frac{n-1}{n}, \dots \right\}$$

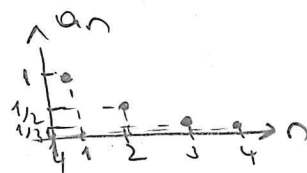
$$\{d_n\} = \{1, -1, 1, -1, \dots, (-1)^{n+1}, \dots\}$$

→ Graphical representation

$$a_n = \frac{1}{n}$$

$$a_4 = 1/4, a_2 = 1/2$$

$$a_3 = 1/3, 1 = a_1$$



Definition (Convergence and Divergence)

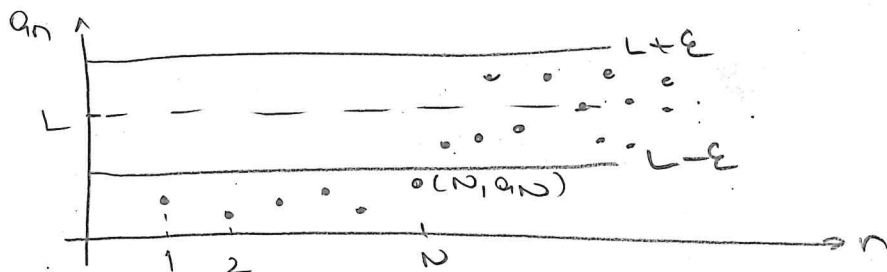
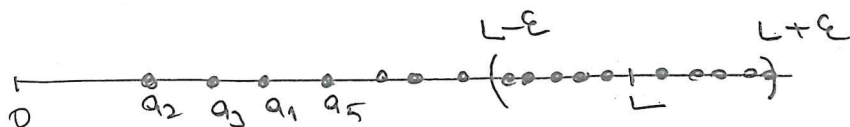
The sequence $\{a_n\}$ converges to the number L if for every positive number ε there corresponds an integer N such that for all n

$$n > N \Rightarrow |a_n - L| < \varepsilon$$

→ If no such number L exists, we say that $\{a_n\}$ diverges.

→ If $\{a_n\}$ converges to L , we write $\lim_{n \rightarrow \infty} a_n = L$, or simply

$a_n \rightarrow L$ and call L the limit of the sequence.



Ex 7

(a) $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ (b) $\lim_{n \rightarrow \infty} k = k$

(c) Show that the sequence $\{1, -1, 1, \dots, (-1)^{n+1}, \dots\}$ diverges

(a) Let $\epsilon > 0$ be given we must find an integer N such that for all $n > N \Rightarrow |\frac{1}{n} - 0| < \epsilon$. $\frac{1}{n} < \epsilon \Rightarrow n > 1/\epsilon$. If N is any integer greater than $1/\epsilon$ it will hold $n > N$.

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

(b) Let $\epsilon > 0$, Find N such that $n > N \Rightarrow |k - k| < \epsilon$ for any positive integer it holds.

(c) Suppose there is a number L that sequence converges.

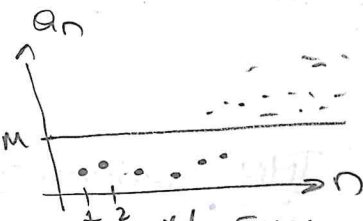
Choose $\epsilon = 1/2$ it follows that $|L - L| < 1/2$ $1/2 < L < 3/2$

Similarly for -1 we have $|L + 1| < 1/2$ $-3/2 < L < -1/2$

L values lie in different intervals. Therefore, we say that limit does not exist.

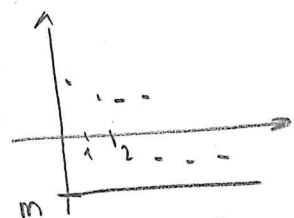
DEFINITION The sequence $\{a_n\}$ diverges to infinity if for every number M there is an integer N such that for all n larger than N , $a_n > M$. If this condition holds we write

$$\lim_{n \rightarrow \infty} a_n = \infty \quad \text{or} \quad a_n \rightarrow \infty$$



Similarly if for every m there is an integer N such that for all $n > N$ we have $a_n < m$, then we say $\{a_n\}$ diverges to negative infinity and write

$$\lim_{n \rightarrow \infty} a_n = -\infty \quad \text{or} \quad a_n \rightarrow -\infty$$



THEOREM (Calculating Limits of Sequences)

Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers, and let A and B be real numbers. The following rules hold if $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$

- ① Sum rule $\lim_{n \rightarrow \infty} (a_n + b_n) = A + B$
- ② Difference Rule $\lim_{n \rightarrow \infty} (a_n - b_n) = A - B$
- ③ Constant Multiple Rule $\lim_{n \rightarrow \infty} (k \cdot b_n) = k \cdot A \quad k \in \mathbb{R}$
- ④ Product Rule $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = A \cdot B$
- ⑤ Quotient Rule $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B} \quad \text{if } B \neq 0$

Ex 7 ① $\lim_{n \rightarrow \infty} \left(-\frac{1}{n}\right) = -1 \cdot \lim_{n \rightarrow \infty} \frac{1}{n} = -1 \cdot 0 = 0$

② $\lim_{n \rightarrow \infty} \left(\frac{n-1}{n}\right) = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) = 1$

③ $\lim_{n \rightarrow \infty} \frac{5}{n^2} = 5 \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$

④ $\lim_{n \rightarrow \infty} \frac{4 - 7n^6}{n^6 + 3} = \lim_{n \rightarrow \infty} \frac{n^6(4/n^6 - 7)}{n^6(1 + 3/n^6)} = -7$

THEM Sandwich Theorem for Sequences. Let $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$

be sequences of real numbers. If $a_n \leq b_n \leq c_n$ holds for all n beyond some index N , and if $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$

then $\lim_{n \rightarrow \infty} b_n = L$ also

Ex 7 (a) $\lim_{n \rightarrow \infty} \frac{\cos n}{n} = ?$

$$\lim_{n \rightarrow \infty} \frac{\cos n}{n} = 0$$

$$-1 \leq \cos n \leq 1 \quad -\frac{1}{n} \leq \frac{\cos n}{n} \leq \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} -\frac{1}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

S.T

(b) $\lim_{n \rightarrow \infty} \frac{1}{2^n} = ?$

$$\lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$$

$$0 \leq \frac{1}{2^n} \leq \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} 0 = 0 \quad \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad \text{S.T}$$

(c) $\lim_{n \rightarrow \infty} (-1)^n \frac{1}{n} = ?$

$$\lim_{n \rightarrow \infty} (-1)^n \frac{1}{n} = 0$$

$$-\frac{1}{n} \leq (-1)^n \frac{1}{n} \leq \frac{1}{n}$$

$$\text{S.T} \quad \lim_{n \rightarrow \infty} -\frac{1}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

THM The Continuous Function Theorem for sequences.

Let $\{a_n\}$ be a sequence of real numbers. If $a_n \rightarrow L$ and if f is a function that is continuous at L and defined at all a_n , then $f(a_n) \rightarrow f(L)$

Ex 7 Show that $\sqrt{(n+1)/n} \rightarrow 1$

We know that $\lim_{n \rightarrow \infty} \frac{n+1}{n} = 1$. Since function $y = \sqrt{x}$ is

continuous, we have $\sqrt{\frac{n+1}{n}} \rightarrow 1$

Ex 7 $\lim_{n \rightarrow \infty} 2^{1/n} = ?$

$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$. $f(x) = 2^x$ is a continuous function. Therefore

$$\lim_{n \rightarrow \infty} 2^{1/n} = 2^{\lim_{n \rightarrow \infty} 1/n} = 2^0 = 1$$

Theorem If $\lim_{n \rightarrow \infty} |a_n| = 0$, then $\lim_{n \rightarrow \infty} a_n = 0$

Ex 7 $\lim_{n \rightarrow \infty} \left| \frac{(-1)^n}{n} \right| = ?$

(7) $\lim_{n \rightarrow \infty} \left| \frac{(-1)^n}{n} \right| = 0$ so $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

Thm Suppose that $f(x)$ is a function defined for all $x \geq n_0$ and that $\{a_n\}$ is a sequence of real numbers such that $a_n = f(n)$ for $n \geq n_0$. Then

$$\lim_{x \rightarrow \infty} f(x) = L \Rightarrow \lim_{n \rightarrow \infty} a_n = L$$

Ex 7 Show that $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$

Let $a_n = \frac{\ln n}{n}$. If $f(x) = \frac{\ln x}{x}$, then $f(x)$ is a function for $x \geq 1$ and $a_n = f(n)$. Thus,

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = \lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0$$

L.H.

Ex 7 Does the sequence whose n th term is $a_n = \left(\frac{n+1}{n-1}\right)^n$ converge. If so, $\lim_{n \rightarrow \infty} a_n = ?$

$$\lim_{n \rightarrow \infty} \left(\frac{n+1}{n-1}\right)^n = (1^\infty)$$

Use L.H

$$a_n = \left(\frac{n+1}{n-1}\right)^n \Rightarrow \ln a_n = n \ln \left(\frac{n+1}{n-1}\right)$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \ln a_n &= \lim_{n \rightarrow \infty} n \ln \left(\frac{n+1}{n-1}\right) \\ &= \lim_{n \rightarrow \infty} \frac{\ln \left(\frac{n+1}{n-1}\right)}{\frac{1}{n}} \quad \left(\frac{0}{0}\right) \\ &= \lim_{n \rightarrow \infty} \frac{\frac{(n+1) - (n-1)}{(n-1)^2}}{\frac{-1}{n^2}} \\ &= \lim_{n \rightarrow \infty} \frac{\frac{2}{(n-1)^2}}{-\frac{1}{n^2}} \end{aligned}$$

$$\lim_{n \rightarrow \infty} \ln a_n = 2 \quad \lim_{n \rightarrow \infty} a_n = e^2$$

(1)

THIR Commonly Occurring Limits

→ The following six sequences converges to the limits listed below

① $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$

② $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$

③ $\lim_{n \rightarrow \infty} x^{1/n} = 1 \quad (x \neq 0)$

④ $\lim_{n \rightarrow \infty} x^n = 0$

⑤ $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$
($x \in \mathbb{R}$)

⑥ $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0 \quad (x \in \mathbb{R})$

Ex 7 a) $\lim_{n \rightarrow \infty} \frac{\ln(n^2)}{n} = ? \quad \lim_{n \rightarrow \infty} \frac{\ln n^2}{n} = \lim_{n \rightarrow \infty} 2 \frac{\ln n}{n} = 0$

b) $\lim_{n \rightarrow \infty} \sqrt[n]{n^2} = ?$

$\lim_{n \rightarrow \infty} n^{2/n} = \lim_{n \rightarrow \infty} (n^{1/n})^2 = 1^2 = 1$

c) $\lim_{n \rightarrow \infty} \sqrt[n]{3n} = ?$

$\lim_{n \rightarrow \infty} \sqrt[n]{3n} = \lim_{n \rightarrow \infty} (3n)^{1/n} = \lim_{n \rightarrow \infty} 3^{1/n} \cdot n^{1/n} = 1$

d) $\lim_{n \rightarrow \infty} \left(-\frac{1}{2}\right)^n = ?$

$\lim_{n \rightarrow \infty} \left(-\frac{1}{2}\right)^n = 0$ (rule ④) $x = -1/2$

e) $\lim_{n \rightarrow \infty} \left(\frac{n-2}{n}\right)^n = \lim_{n \rightarrow \infty} \left(1 - \frac{2}{n}\right)^n = e^{-2}$

f) $\lim_{n \rightarrow \infty} \frac{100^n}{n!} = ?$ (rule ⑥) $x = 100$ $\lim_{n \rightarrow \infty} \frac{100^n}{n!} = 0$

Bounded Monotonic Sequences

Definitions

A sequence $\{a_n\}$ is bounded from above if there exists a number M such that $a_n \leq M$ for all n . The number M is an upper bound for $\{a_n\}$. If M is an upper bound for $\{a_n\}$ but no number less than M is an upper bound for $\{a_n\}$, then M is the least upper bound for $\{a_n\}$.

($a_n \leq M$ upper bound

There is no $K < M$ such that $a_n \leq K$ then M

least upper bound.

→ A sequence $\{a_n\}$ is bounded from below if there exists a number m such that $a_n \geq m$ for all n . The number m is a lower bound for $\{a_n\}$. If m is a lower bound for $\{a_n\}$ but no number greater than m is a lower bound for $\{a_n\}$, then m is the greatest lower bound for $\{a_n\}$.

($a_n \geq m$
 $\downarrow$
 lower bound)
 $a_n \geq m$: There is no k such that $k > m$ and $a_n \geq k$ then m is greatest lower bound.

→ If $\{a_n\}$ is bounded from above and below, the $\{a_n\}$ is bounded. If $\{a_n\}$ is not bounded, we say that $\{a_n\}$ is an unbounded sequence.

Bounded Monotonic Sequence

Ex ① The sequence $1, 2, 3, \dots, n, \dots$ has no upper bound. But the number $m = 1$ greatest lower bound.

② The sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots, \frac{n}{n+1}, \dots$ is bounded above by every real number greater or equal to 1. The upper bound $M = 1$. The sequence is also bounded below by every number less than or equal to $\frac{1}{2}$, which is its greatest lower bound.

Definition A sequence $\{a_n\}$ is nondecreasing if $a_n \leq a_{n+1}$ for all n . That is, $a_1 \leq a_2 \leq a_3 \leq \dots$. The sequence is nonincreasing if $a_n \geq a_{n+1}$ for all n . The sequence $\{a_n\}$ is monotonic if it is either nondecreasing or nonincreasing.

- Ex
- (a) The sequence $1, 2, 3, \dots, n, \dots$ is nondecreasing
 - (b) The sequence $1/2, 2/3, 3/4, \dots, 1/n+1$ is nondecreasing
 - (c) The sequence $1, 1/2, 1/4, 1/8, \dots, 1/2^n, \dots$ is nonincreasing
 - (d) The sequence $3, 3, 3, \dots, 3, \dots$ is both nondecreasing and nonincreasing
 - (e) The sequence $1, -1, 1, -1, 1, -1, \dots$ is not monotonic

THM The Monotonic Sequence Theorem

If a sequence $\{a_n\}$ is both bounded and monotonic, then the sequence converges.

Ex Let a_n be defined recursively by

$$a_1 = 1 \quad a_{n+1} = \sqrt{6 + a_n} \quad (n = 1, 2, 3, \dots)$$

Show that $\lim_{n \rightarrow \infty} a_n$ exists and find its value.

Solution Observe that $a_2 = \sqrt{6+1} = \sqrt{7} > a_1$. If $a_{k+1} > a_k$

then $a_{k+2} = \sqrt{6+a_{k+1}} > \sqrt{6+a_k} = a_{k+1}$ so $\{a_n\}$ increasing,

by induction. Now observe that $a_1 = 1 < 3$. If $a_k < 3$ then

$$a_{k+1} = \sqrt{6+a_k} < \sqrt{6+3} = 3 \quad \text{so } a_{k+1} < 3 \quad \text{so for every } n$$

$a_n < 3$. $\{a_n\}$ is increasing and bounded above $\lim_{n \rightarrow \infty} a_n = L$

Since $\sqrt{6+x}$ is continuous function of x

$$L = \lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \sqrt{6+a_n} = \sqrt{\lim_{n \rightarrow \infty} (6+a_n)} = \sqrt{6+L}$$

$$L = \sqrt{6+L}$$

$$L^2 = 6+L \quad (L-3)(L+2) = 0 \Rightarrow L = 3 \text{ or } L = -2$$

Since $a_n \geq 1$ for all n , therefore $\lim_{n \rightarrow \infty} a_n = 3$

SERIES

Definition Given a sequence of numbers $\{a_n\}$, an expression of the form

$$a_1 + a_2 + a_3 + \dots + a_n + \dots$$

is an infinite series. The number a_n is the n th term of the series. The sequence $\{s_n\}$ is defined by

$$s_1 = a_1$$

$$s_2 = a_1 + a_2$$

$$s_3 = a_1 + a_2 + a_3$$

$$s_n = a_1 + a_2 + a_3 + \dots + a_n = \sum_{k=1}^n a_k$$

is the sequence of partial sums of the series, the number s_n being the n th partial sum. If the sequence of partial sums converges to a limit L , we say that the series converges and that its sum is L . In this case we also write

$$a_1 + a_2 + \dots + a_n + \dots = \sum_{n=1}^{\infty} a_n = L$$

If the sequence of partial sums of the series do not converge, we say that the series diverges.

Ex 7 $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots = ?$

	<u>Value</u>	<u>Suggestive expression</u>
$s_1 = 1$	1	$2 - 1$
$s_2 = 1 + \frac{1}{2}$	$\frac{3}{2}$	$2 - \frac{1}{2}$
$s_3 = 1 + \frac{1}{2} + \frac{1}{4}$	$\frac{7}{4}$	$2 - \frac{1}{4}$

$$s_n = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^{n-1}} = \frac{2^n - 1}{2^{n-1}} \quad 2 - \frac{1}{2^{n-1}}$$

The sum of the infinite series $1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^{n-1}} + \dots$ is 2.

Geometric Series

Geometric series are series of the form

$$a + ar + ar^2 + \dots + ar^{n-1} + \dots = \sum_{n=1}^{\infty} ar^{n-1} = \sum_{n=0}^{\infty} ar^n$$

in which a and r are fixed real numbers and $a \neq 0$. The ratio r can be positive, as in

$$1 + \frac{1}{2} + \frac{1}{4} + \dots + \left(\frac{1}{2}\right)^{n-1} + \dots \quad r = \frac{1}{2}, a = 1$$

or negative, as in

$$1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \dots + \left(-\frac{1}{3}\right)^{n-1} + \dots \quad r = -\frac{1}{3}, a = 1$$

→ If $r = 1$, the n th partial sum of the geometric series is

$$S_n = a + a(1) + a(1)^2 + a(1)^3 + \dots + a(1)^{n-1} = n \cdot a$$

and the series diverges because $\lim_{n \rightarrow \infty} S_n = \pm \infty$, depending on

the sign of a . If $r = -1$, the series diverges because the partial sum alternate between a and 0 .

→ If $|r| = 1$,

$$S_n = a + ar + ar^2 + \dots + ar^{n-1}$$

$$rS_n = ar + ar^2 + \dots + ar^{n-1} + ar^n$$

$$S_n - rS_n = a - ar^n$$

$$S_n(1-r) = a(1-r^n)$$

$$S_n = \frac{a(1-r^n)}{(1-r)} \quad (r \neq 1)$$

If $|r| < 1$, the geometric series $a + ar + ar^2 + \dots + ar^{n-1} + \dots$ converges to $a/(1-r)$

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r} \quad |r| < 1$$

If $|r| \geq 1$, the series diverges.

Ex 9 The geometric series with $a = 1/9$ and $r = 1/3$ is

$$\frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \dots = \sum_{n=1}^{\infty} \frac{1}{9} \left(\frac{1}{3}\right)^{n-1} = \frac{1/9}{1-1/3} = \frac{1}{6}$$

(b) The series $\sum_{n=0}^{\infty} \frac{(-1)^n 5}{4^n} = 5 - \frac{5}{4} + \frac{5}{16} - \frac{5}{64} + \dots$

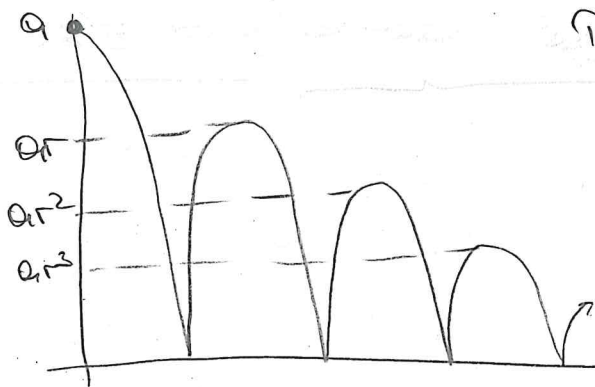
is a geometric series with $a = 5$ and $r = -1/4$. It converges to

$$\frac{a}{1-r} = \frac{5}{1+1/4} = 4$$

(c) You drop a ball from a meters above a flat surface.

Each ball hits the surface after falling a distance h , it

rebounds a distance rh , where r is positive but less than 1. Find the total distance the ball travels up and down



The total distance $\frac{2ar}{1-r}$

$$s = a + 2ar + 2ar^2 + 2ar^3 + \dots =$$

$$= a + \frac{2ar}{1-r} = a \frac{1+r}{1-r}$$

If $a = 6m$ and $r = 2/3$ then distance $s = 6 \cdot \frac{1+2/3}{1-2/3} = 30m$.

$$2ar + 2ar^2 + 2ar^3 + \dots = \sum_{n=1}^{\infty} 2ar^{n-1} = 2a$$

$$= 2a \frac{1}{1-r} - 2a = \frac{2a(1-(1-r))}{1-r}$$

$$= \frac{2ar}{1-r}$$

d) Express the repeating decimal $5.232323 \dots$ as the ratio of two integers.

From the definition of a decimal number, we get a geometric series

$$\begin{aligned} 5.\underline{23}\underline{23}\underline{23} &= 5 + \frac{23}{100} + \frac{23}{(100)^2} + \frac{23}{(100)^3} + \dots \\ &= 5 + \frac{23}{100} \left[1 + \frac{1}{100} + \frac{1}{100^2} + \frac{1}{100^3} + \dots \right] \quad \begin{matrix} a=1 \\ r=1/100 \end{matrix} \\ &\quad \frac{1}{1-1/100} \\ &= 5 + \frac{23}{100} \left(\frac{1}{0.99} \right) = 5 + \frac{23}{99} = \frac{515}{99} \end{aligned}$$

Ex Find the sum of the "telescoping" series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

$$\sum_{n=1}^k \frac{1}{n(n+1)} = \sum_{n=1}^k \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

$$S_k = \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{k} - \frac{1}{k+1} \right)$$

$$S_k = 1 - \frac{1}{k+1} \quad \lim_{k \rightarrow \infty} S_k = \lim_{k \rightarrow \infty} 1 - \frac{1}{k+1} = 1$$

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$$

The n-th Term Test For a Divergent Series

THEOREM (TEST I)

If $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \rightarrow 0$.

The n-th Term Test for Divergence (TEST I)

$\sum_{n=1}^{\infty} a_n$ diverges, if $\lim_{n \rightarrow \infty} a_n$ fails to exist or differs from zero.

Ex 7 (a) $\sum_{n=1}^{\infty} n^2$ since $\lim_{n \rightarrow \infty} n^2 = \infty$ the series is divergent

(b) $\sum_{n=1}^{\infty} \frac{n+1}{n}$ $\lim_{n \rightarrow \infty} \frac{n+1}{n} = 1$ diverges. ($\lim_{n \rightarrow \infty} a_n \neq 0$)

(c) $\sum_{n=1}^{\infty} (-1)^{n+1}$ $\lim_{n \rightarrow \infty} (-1)^{n+1}$ doesn't exist

(d) $\sum_{n=1}^{\infty} \frac{-1}{2n+5}$ $\lim_{n \rightarrow \infty} \frac{-1}{2n+5} = -\frac{1}{2} \neq 0$

COMBINING SERIES

THM If $\sum a_n = A$ and $\sum b_n = B$ are convergent series, then

(1) Sum rule $\sum (a_n + b_n) = \sum a_n + \sum b_n = A + B$

(2) Difference Rule $\sum (a_n - b_n) = \sum a_n - \sum b_n = A - B$

(3) Constant Multiple Rule $\sum k a_n = k \sum a_n = k A$ (any $k \in \mathbb{R}$)

(*) Every nonzero constant multiple of a divergent series diverges

(*) If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum (a_n + b_n)$ and $\sum (a_n - b_n)$ both diverge.

Caution !!!

Remember that $\sum (a_n + b_n)$ can converge when $\sum a_n$ and $\sum b_n$ both diverge. For example,

$$\sum a_n = 1 + 1 + 1 + \dots$$

$$\sum b_n = (-1) + (-1) + (-1) + \dots$$

diverge whereas $\sum (a_n + b_n) = 0 + 0 + 0 + \dots$ converges to 0

Ex $\sum_{n=1}^{\infty} \frac{3^{n-1} - 1}{6^{n-1}} = \sum_{n=1}^{\infty} \left(\frac{1}{2^{n-1}} - \frac{1}{6^{n-1}} \right) = \sum_{n=1}^{\infty} \frac{1}{2^{n-1}} - \sum_{n=1}^{\infty} \frac{1}{6^{n-1}}$

$$= \frac{1}{1 - 1/2} - \frac{1}{1 - 1/6} = 2 - \frac{6}{5} = \frac{4}{5}$$

Adding or Deleting Terms

→ we can add a finite number of terms to a series or delete a finite number of terms without altering the series' converge or diverge, although in the case of converge this will usually change the sum. If $\sum_{n=1}^{\infty} a_n$ converges

then $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \dots + a_{k-1} + \sum_{n=k}^{\infty} a_n$$

Conversely if $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$, then $\sum_{n=1}^{\infty} a_n$ converges

$$\sum_{n=1}^{\infty} \frac{1}{5^n} = \frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \sum_{n=4}^{\infty} \frac{1}{5^n}$$

$$\sum_{n=4}^{\infty} \frac{1}{5^n} = \sum_{n=1}^{\infty} \frac{1}{5^n} - \frac{1}{5} - \frac{1}{25} - \frac{1}{125}$$

Reindexing

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1+h}^{\infty} a_{n-h} = a_1 + a_2 + a_3 + \dots$$

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1-h}^{\infty} a_{n+h} = a_1 + a_2 + a_3 + \dots$$

Ex 7 We can write the geometric series

$$\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 1 + \frac{1}{2} + \frac{1}{4} + \dots$$

$$\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = \sum_{n=0}^{\infty} \frac{1}{2^n} = \sum_{n=5}^{\infty} \frac{1}{2^{n-5}} = \sum_{n=-4}^{\infty} \frac{1}{2^{n+4}}$$

Corollary A series $\sum_{n=1}^{\infty} a_n$ of nonnegative terms converges

if and only if its partial sums are bounded above.

Ex 7 The series (HARMONIC SERIES #)

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} + \dots$$

is called harmonic series.

$$\begin{array}{c} S_{23} > 3/2 \\ \hline 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \left(\frac{1}{9} + \frac{1}{10} + \dots + \frac{1}{16}\right) \\ \qquad \qquad \qquad > \frac{2}{4} = \frac{1}{2} \qquad \qquad \qquad > \frac{4}{8} = \frac{1}{2} \qquad \qquad \qquad > \frac{8}{16} = \frac{1}{2} \end{array}$$

$$S_{2^k} > \frac{k}{2} \text{ for all } k \geq 1 \Rightarrow \sum \frac{1}{n} \text{ diverges.}$$

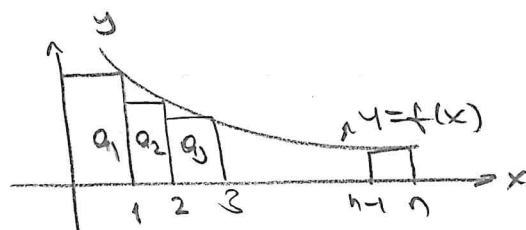
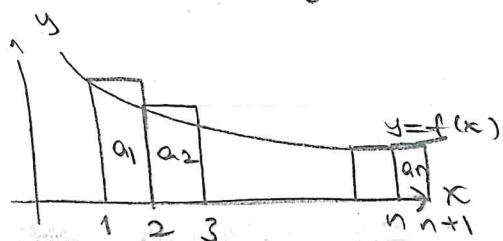
THE INTEGRAL TEST

THM - THE INTEGRAL TEST (TEST II)

Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is continuous, positive, decreasing function of x for all $x \geq N$ (N a positive integer)

Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(x) dx$

both converge or both diverge



Ex

Show that the series (p-test)

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \dots + \frac{1}{n^p} + \dots$$

(p a real constant) converges if $p > 1$, and diverges if $p \leq 1$

Sol If $p > 1$, then $f(x) = \frac{1}{x^p}$ is a positive decreasing function of x . Since

$$\begin{aligned} \int_1^{\infty} \frac{1}{x^p} dx &= \int_1^{\infty} x^{-p} dx = \lim_{b \rightarrow \infty} \int_1^b x^{-p} dx = \lim_{b \rightarrow \infty} \left. \frac{x^{-p+1}}{-p+1} \right|_1^b \\ &= \frac{1}{1-p} \lim_{b \rightarrow \infty} \left(\frac{1}{b^{p-1}} - 1 \right) \quad \begin{matrix} b \rightarrow \infty & \frac{1}{b^{p-1}} \rightarrow 0 \\ p-1 > 0 \end{matrix} \\ &= \frac{1}{1-p} (0 - (-1)) = \frac{1}{p-1} \end{aligned}$$

the series converges by the Integral Test
(12)

Ex 7 The series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ is not a p-series, but it converges by the integral test.

The function $f(x) = \frac{1}{x^2+1}$ is positive, continuous and decreasing

for $x \geq 1$

$$\int_1^{\infty} \frac{1}{1+x^2} dx = \lim_{b \rightarrow \infty} [\arctan x]_1^b = \lim_{b \rightarrow \infty} [\arctan b - \arctan 1] = \pi/2 - \pi/4 = \pi/4$$

COMPARISON TESTS

THM The Comparison Test. (TEST II)

Let $\sum a_n$, $\sum c_n$ and $\sum d_n$ be series with nonnegative terms. Suppose that for some integer N

$$d_n \leq a_n \leq c_n \text{ for all } n > N$$

(a) If $\sum c_n$ converges, then $\sum a_n$ also converges

(b) If $\sum d_n$ diverges, then $\sum a_n$ also diverges

Ex 7 The series $\sum_{n=1}^{\infty} \frac{5}{5n-1}$ diverges because the n th term

$$\frac{5}{5n-1} = \frac{1}{n-\frac{1}{5}} > \frac{1}{n} \text{ is greater than the } n\text{th term}$$

of the divergent harmonic series.

Ex 7 $\sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots$ converges or diverges?

It's all terms are positive.

$$1 + \sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots$$

(*) It doesn't mean $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges to 3.
 This series is convergent $1 + \sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + \frac{1}{1-1/2} = 3$
 Since, each term of the series $\sum_{n=0}^{\infty} \frac{1}{n!}$ is less than or equal to $\frac{1}{2^n}$ (which is important and it

Ex The series

$$5 + \frac{2}{3} + \frac{1}{7} + 1 + \frac{1}{2+\sqrt{1}} + \frac{1}{4+\sqrt{2}} + \frac{1}{5+\sqrt{3}} + \dots + \frac{1}{2^{\frac{1}{2}}+\sqrt{n}}$$

Converges

$$1 + \frac{1}{2+\sqrt{1}} + \frac{1}{4+\sqrt{2}} + \frac{1}{5+\sqrt{3}} + \dots \leq 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$$

$$\sum_{n=1}^{\infty} r^{n-1} = \frac{1}{1-r} = 2$$

Therefore, left side converges.

$$\sum_{n=0}^{\infty} \frac{1}{2^n + \sqrt{n}} \Rightarrow \text{converges}$$

$\underbrace{\hspace{1cm}}_{a_n}$

$$\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n-1} \Rightarrow \text{converges}$$

$\underbrace{\hspace{1cm}}_{b_n}$

$a_n \leq c b_n$ comparison test
gives $\sum_{n=0}^{\infty} a_n$ converges.

THM The Limit Comparison Test (TEST IV)

Suppose that $a_n > 0$ and $b_n > 0$ for all $n \geq N$ (N an integer)

① If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge

② If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges

③ If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Ex Which of the following series converge, and which diverge?

① $\frac{3}{4} + \frac{5}{9} + \frac{7}{16} + \dots = \sum_{n=1}^{\infty} \frac{2n+1}{(n+1)^2} = \sum_{n=1}^{\infty} \frac{2n+1}{n^2+2n+1}$

② $\frac{1}{1} + \frac{1}{5} + \frac{1}{7} + \frac{1}{15} + \dots = \sum_{n=1}^{\infty} \frac{1}{2^n - 1}$

③ $\frac{1+2\ln 2}{9} + \frac{1+3\ln 3}{14} + \frac{1+4\ln 4}{21} + \dots = \sum_{n=2}^{\infty} \frac{1+n\ln n}{n^2+n}$

Solution ① $a_n = \frac{2n+1}{n^2+2n+1}$ we expect a_n behaves like

$\frac{2n}{n^2} = \frac{2}{n}$ since the leading term n^2 dominates for large n

we let $b_n = \frac{1}{n}$. Since, $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n}$ diverges
 harmonic series
 and $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{2n+1}{\frac{n^2+2n+1}{1/n}} = \lim_{n \rightarrow \infty} \frac{2n^2+n}{n^2+2n+1} = 2$
 $\rightarrow \sum a_n$ diverges.

(b) Let $a_n = \frac{1}{2^n - 1}$. For large n , we expect a_n to
 behave like $\frac{1}{2^n}$, so we let $b_n = \frac{1}{2^n}$
 $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{2^n}$ converges
 geometric series
 $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{2^n - 1}}{\frac{1}{2^n}} = \lim_{n \rightarrow \infty} \frac{2^n}{2^n - 1} = \lim_{n \rightarrow \infty} \frac{1}{1 - \frac{1}{2^n}} = 1$
 $\sum_{n=1}^{\infty} a_n$ converges

(c) Let $a_n = \frac{1+n \ln n}{n^2+5}$. For large n , we expect a_n to
 behave like $\frac{n \ln n}{n^2} = \frac{\ln n}{n} \geq \frac{1}{n}$ for $n \geq 3$
 $\sum_{n=2}^{\infty} b_n = \sum_{n=2}^{\infty} \frac{1}{n}$ diverges
 p-series, harmonic series
 $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n+n^2 \ln n}{n^2+5} = \infty$
 $\sum a_n$ diverges.

THE RATIO AND ROOT TESTS

THM: THE RATIO TEST (TEST V)

Let $\sum a_n$ be a series with positive terms and

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \rho$$

Then (a) the series are convergent if $\rho < 1$

(b) the series are divergent if $\rho > 1$ or infinite

(c) the test is inconclusive if $\rho = 1$

Ex Investigate the convergence of the following series.

(a) $\sum_{n=0}^{\infty} \frac{2^n + 5}{3^n}$

(b) $\sum_{n=1}^{\infty} \frac{(2n)!}{n! \cdot n!}$

(a)
$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1} + 5}{3^{n+1}} \cdot \frac{3^n}{2^n + 5} = \frac{1}{3} \cdot \frac{2^{n+1} + 5}{2^n + 5} = \frac{1}{3} \left(\frac{2 + 5 \cdot 2^{-n}}{1 + 5 \cdot 2^{-n}} \right)$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{1}{3} \lim_{n \rightarrow \infty} \frac{2 + 5 \cdot 2^{-n}}{1 + 5 \cdot 2^{-n}} = \frac{2}{3} < 1 \text{ series is convergent}$$

(*) This doesn't MEAN the sum equals to $2/3$

$$\sum_{n=0}^{\infty} \frac{2^n + 5}{3^n} = \sum_{n=0}^{\infty} \left(\frac{2}{3} \right)^n + 5 \cdot \sum_{n=0}^{\infty} \frac{1}{3^n} = \frac{1}{1 - 2/3} + \frac{5}{1 - 1/3} = \frac{21}{2}$$

(b)
$$\frac{a_{n+1}}{a_n} = \frac{(2n+2)!}{(n+1)! (n+1)!} \cdot \frac{n! \cdot n!}{(2n)!} = \frac{(2n+1)(2n+2)}{(n+1)(n+1)} = \frac{4n+2}{n+1}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{4n+2}{n+1} = 4 > 1 \text{ series is } \underline{\text{divergent}}$$

THM THE ROOT TEST (TEST VI)

Let $\sum a_n$ be series with $a_n \geq 0$ for $n \geq 1$

and suppose that

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \rho$$

- Then:
- (a) the series converges if $\rho < 1$
 - (b) the series diverges if $\rho > 1$ or ρ is infinite
 - (c) The test is inconclusive if $\rho = 1$

Ex which of the following series converge and which diverge

(a) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ (b) $\sum_{n=1}^{\infty} \frac{2^n}{n^3}$ (c) $\sum_{n=1}^{\infty} \left(\frac{1}{1+n}\right)^n$

(a) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ converges because $\sqrt[n]{\frac{n^2}{2^n}} = \frac{n^{2/n}}{2}$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\frac{n^2}{2^n}} = \lim_{n \rightarrow \infty} \frac{n^{2/n}}{2} = \frac{1^2}{2} < 1 \text{ converges}$$

(b) $\lim_{n \rightarrow \infty} \left(\frac{2^n}{n^3}\right)^{1/n} = \lim_{n \rightarrow \infty} \frac{2}{n^{3/n}} = \frac{2}{1^3} = 2 > 1 \text{ diverges}$

(c) $\lim_{n \rightarrow \infty} \left(\frac{1}{1+n}\right)^n = \lim_{n \rightarrow \infty} \frac{1}{1+n} = 0 < 1 \text{ converges}$